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Investigating the accuracy of cross-learning time series forecasting methods

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ABSTRACT

The M4 competition identified innovative forecasting methods, advancing the theory and practice of forecasting. One of the most promising innovations of M4 was the utilization of cross-learning approaches that allow models to learn from multiple series how to accurately predict individual ones. In this paper, we investigate the potential of cross-learning by developing various neural network models that adopt such an approach, and we compare their accuracy to that of traditional models that are trained in a series-by-series fashion. Our empirical evaluation, which is based on the M4 monthly data, confirms that cross-learning is a promising alternative to traditional forecasting, at least when appropriate strategies for extracting information from large, diverse time series data sets are considered. Ways of combining traditional with cross-learning methods are also examined in order to initiate further research in the field.
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1. Introduction

The M4 competition (Makridakis, Spiliotis et al., 2020) evaluated a variety of forecasting methods using a large, diverse set of 100,000 time series (Spiliotis, Kouloumos et al., 2020). The competition caught the attention of the forecasting community by identifying innovative forecasting methods, setting new, high standards in the quality of forecasting research (Hyndman, 2020). It was entirely open-source (The M4 Team, 2018) and transparent (Makridakis, Assimakopoulos et al., 2018). In this respect, researchers and practitioners are expected to be inspired to explore new, promising forecasting approaches to further advance the theory and practice of forecasting (Petropoulos & Makridakis, 2020b). This is also the objective of the present paper, which tries to shed some light on the accuracy and the potential of cross-learning (CL) forecasting approaches, i.e., models that are trained across an entire time series data set in order to extract information from multiple series and accurately predict individual ones.

CL is one of the most innovative and promising approaches successfully implemented at the M4 competition, as it allows for various patterns observed along mostly unrelated series to be effectively learned, thus mitigating the limitations of traditional forecasting methods trained in a series-by-series fashion (Makridakis, Spiliotis, Assimakopoulos, 2018). This is justified by observing that the top three performing methods of the competition used information from the complete data set to predict each separate series (Makridakis, Spiliotis et al., 2020). However, further investigation is required to conclude whether such information was responsible for the success of these methods, and to what extent. This is exactly the motivation for the research conducted in this study.

Before proceeding with the analysis, it is important to clarify the differences between traditional and CL forecasting models. Simply put, traditional models are trained using only the data of the single series they are going to predict. This means that N models are required

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for predicting N different series. Statistical models—such as exponential smoothing (Hyndman et al., 2002) and ARIMA (Hyndman & Khandakar, 2008), as well as the winning submission of the M3 competition (Makridakis & Hibon, 2000), Theta (Assimakopoulos & Nikolopoulos, 2000)—can therefore be classified as traditional. Similarly, machine learning (ML) models, such as the two ML benchmarks of M4, which use multiple windows from a single series for training, also fall into this category. On the other hand, a model would be characterized as CL when it is trained using data from multiple series without focusing only on the observations originating from the particular series it is going to predict. In this case, a single CL model can be trained on all N time series of a data set and then used to forecast them as well. In the M4 competition, a variety of CL methods were introduced, with varying degrees of success (Makridakis, Spiliotis et al., 2020).

Another interesting class of methods that takes advantage of both traditional and CL modeling approaches is the "hybrid". In the M4 competition, Montero-Manso et al. (2020) (Second Place) introduced a hybrid time series feature-based (FB) method, which utilized an XGBoost model trained across the complete M4 data set to determine the optimal weights for combining eight different traditional models trained per series that provided the base forecasts of the combination scheme. Similarly, Smyl (2020) (First Place) submitted a method that utilized a dynamic computational graph neural network (NN) system to mix standard exponential smoothing with advanced long short-term memory networks into a common framework. The excellent performance of these methods indicates that there are several possible ways of developing CL forecasting models or combining them with traditional ones in order to improve forecasting performance:

- Directly combining (averaging) the forecasts produced by a CL model and a model trained in a series-by-series fashion.
- Combining the forecasts produced by traditional models based on information extracted from the entire time series data set using FB approaches.
- Adjusting the forecasts produced by a CL model based on the output of a series-specific model.

Despite their excellent performance, CL and hybrid methods remain mostly unexplored, with the majority of studies focusing on the development of FB meta-learning forecasting approaches for selecting or combining traditional models (Lemke & Gabrys, 2010; Talagala et al., 2018; Wang et al., 2009). Only a few studies pertain to the construction of true hybrids (Smyl, 2017). Thus, it is important to evaluate the potential of CL and hybrid methods over standard benchmarks, both statistical and ML ones, before developing highly customized and complex models. This is the goal of the present study, which aims to explore various CL and hybrid forecasting approaches, discuss their advantages and limitations, and by doing so, assist researchers and practitioners in identifying best practices in time series forecasting.

The rest of the paper is structured as follows: Section 2 discusses the advantages of both CL and traditional methods, while highlighting their weaknesses and limitations. Section 3 presents the experimental design of this study, including the data set considered for the empirical evaluation, the performance measures utilized, and the benchmarks introduced. Then, the accuracy of CL methods is explored in two different cases. First, Section 4 introduces methods that exploit time series features both as additional NN inputs and as a way of organizing the training process of the networks in an effort to facilitate learning across diverse series of different characteristics. Second, Section 5 introduces hybrid models that combine the information extracted from both the entire data set and each series individually. Finally, Section 6 presents and discusses the results, and Section 7 concludes the paper and proposes fertile areas for further research.

2. Cross-learning and traditional forecasting methods: A discussion

Models trained in a CL or traditional series-by-series fashion have both merits and limitations. However, in real life, selecting the "best" method is not as trivial as identifying the winning submission of a forecasting competition. Depending on the available resources and data at hand, as well as the task's objectives, the most appropriate forecasting method may differ. Moreover, such realizations contribute to reasonably designed forecasting methods and adequately interpreted results.

2.1. Traditional methods

Nowadays, forecasting models that are trained in a series-by-series fashion, both statistical and ML ones, dominate the field of forecasting. This preference can be attributed to many different factors related to the nature and the historical success of these models, as well as to advances in data and computer resources availability over the years (Makridakis, Hyndman et al., 2020).

The amount of available data is a key factor in the ability of ML models to take full advantage of their capacity and extract information effectively. This is because ML models, in contrast to statistical ones, do not assume an underlying data generation process (DGP) a priori, but approximate it by observing the data directly, thus making them data hungry (Barker, 2020). In this respect, when the amount of data is insufficient for training—and, therefore, for successfully extracting the various patterns of the series, such as seasonality and trend—ML models are doomed to fail (Zhang et al., 1998). Data availability can be achieved in three different ways: (i) by increasing the observations of the individual series, (ii) by increasing the number of series in the data set, and (iii) by increasing both the number of series and their length. The first way requires data sampling to be performed at high frequency levels which, until recently, was the case for only a few types of applications, such as in energy forecasting (Chae et al., 2016; Park et al., 1991), stock market forecasting (Fischer & Krauss, 2018; Moghaddam et al., 2016), and demand forecasting (Dekker et al., 2004; Spiliotis, Makridakis et al., 2020; Zotteri & Kalchschmidt, 2007), where ML models have been adopted with great success.

However, long series alone do not enable the utilization of CL methods. This is exactly where the other two ways of increasing data availability become relevant and make the collection of multiple series of diverse characteristics, as well as their alignment, a prerequisite for effective learning. Also, it should be noted that data availability and the characteristics of the resulting training set may vary based on the specific forecasting task and the practitioner's access to data. For example, in the retail sector, when the task is to forecast the demand for products at an item level, retail companies often have many related and chronologically aligned series available for training ML models. On the other hand, when the task is to generate forecasts at the strategic planning level, it is often the case that fewer series are available for building a suitable training set. Thus, additional data unrelated to the target series have to be gathered for training. As a result, depending on the level of resemblance between the training and the target series, different CL approaches may be more suitable. The present study is focused on generating forecasts for the M4 competition's data set, and is thus more representative of tasks that require gathering large number of unrelated series with diverse characteristics. Unfortunately, even today, creating a large and suitable set of series for applying CL remains a challenging task that requires an extensive search of various databases in order to collect series with varied characteristics, sufficient training samples, and no missing values or structural breaks (Spiliotis, Kouloumos et al., 2020).

Computer power limitations have also influenced the preference of forecasters for simpler models trained separately for each series. In the past, researchers and practitioners had to work with few time series, limited resources, and less sophisticated ML algorithms. In this context, CL was typically less effective than traditional learning approaches, as there were not enough data to learn from and, even when there were, hardware and algorithms could not take full advantage of the data. For instance, graphics processing units (GPUs) only became available in the late 1990s, web services in the early 2000s, and Cloud, Hadoop, Spark, and advanced GPUs in the mid-2010s. Accordingly, although NNs and deep learning were introduced in the 1960s (Ivakhnenko, 1968; Rosenblatt, 1957), it was not until the late 1990s when these algorithms became popular and efficient, thus challenging the accuracy of traditional statistical models. This becomes evident by observing the results of the M3 competition (Makridakis & Hibon, 2000), which included an automated artificial NN that performed averagely in comparison to statistical models.

The focus on statistical models was further reinforced by the empirical findings of popular, open forecasting competitions. For example, the M3 competition established generic forecasting methods, such as the Theta method, that have proven accurate and efficient in predicting numerous series of various characteristics. The NN3 competition (Crone et al., 2011) extended the findings of M3 but focused on ML models in an attempt to investigate whether computational intelligence methods could outperform well-established statistical methods. Novel approaches were tested, with some participants even proposing CL methods (Ilies et al., 2007), and accuracy improvements were documented in favor of ML. Despite that, ML submissions were still outperformed by statistical benchmarks, thus failing to challenge the dominance of the latter.

Based on the limitations and findings discussed above, forecasting models that did not rely on the existence of large sets of series or fast processors became popular, making the utilization of statistical models and their combinations standard practice.

2.2. Cross-learning methods

The idea of using information from multiple series when training forecasting models, both statistical and ML, is not entirely new in the field. For example, the introduction of vector autoregression (VAR) models allowed forecasters to capture the interdependencies between series in order to produce more accurate results. However, in the case of a VAR model, the series used for training cannot be totally unrelated, since the underlying assumption of the model is that each series influences the rest. As a result, such models cannot be effectively applied for forecasting numerous, mostly unrelated series, like the ones of the M4 competition. In the NN3 competition, Ilies et al. (2007) proposed a CL approach that ranked first among the rest of the ML submissions but failed to outperform Theta. The poor performance of CL models in comparison to traditional ones did not allow for the popularity of the former to grow. CL models only recently received acclaim in the field of batch time series forecasting. Nowadays, most CL approaches are implemented using ML algorithms. Thus, our discussion focuses on the advantages and drawbacks of adopting ML models trained in a CL fashion.

The most important argument in favor of CL forecasting methods is that they are built to extract knowledge from multiple series (Gür Ali & Yaman, 2013; Ma & Fildes, 2020). The main principle behind this claim is that, although time series are different, common patterns may occur among them. This idea can be exploited to address two major limitations to forecasting with series-specific ML models: data availability and uncertainty.

First, the information extracted from different series can potentially deal with data limitations that are present when working with a single series. When faced with a limited amount of data points, fitting an ML model using data from a single series is challenging. If the model is unable to capture the characteristics of the series and adjust its parameters accordingly, then it is unlikely to produce accurate results. However, when the model is trained using data from the entire data set, it is more likely to be able to approximate the DGP of the series and capture their primary features, linking them to specific inputs and producing more accurate results. Second, even when more data are available, the DGP may change when looking into different windows of the series. Essentially, even if the model could fit the past of a particular series perfectly, there would be no guarantee that it could successfully predict its future (Tashman, 2000). Exploring

the characteristics of more series provides insights into how the DGP of a series might change in the future. CL methods are designed to overcome these obstacles, thus unlocking the potential of ML in batch time series forecasting.

Another argument in favor of CL methods is that they require less time to forecast multiple series compared to ML methods trained in a series-by-series fashion. Although advances in hardware and parallel processing have reduced the overall duration of such tasks, time is still relevant (Makridakis, Spiliotis et al., 2018; Nikolopoulos & Petropoulos, 2018). By analyzing the process of building, training, and using a trained model for extrapolation, most of the time is typically spent on training. In the case of traditional methods, the whole process is repeated for every series in need of forecasting. Additionally, every new series added to the queue requires a new, dedicated model to be built, trained, and used. On the contrary, using a single model to forecast all series requires the first two parts of the process to be executed just once. The cost of training is higher but it is "paid" only once. An NN model based on CL can be used to produce forecasts for the whole set and for any new series outside the set, within a short period of time.

As expected, CL methods also come with their own limitations, which should be carefully taken into consideration. The first limitation, discussed above, is the requirement for a large set of series to enable meaningful training. In order to be able to extrapolate different series without fitting them individually, the underlying model has to learn as many patterns as possible, and providing large amounts of series during training contributes to that. Essentially, when forecasting a series, accuracy depends on whether the model previously observed an input similar to the one provided. It becomes evident that a diverse, dense set of time series is a key variable for effectively performing this task.

The second major limitation to CL methods lies with the need to apply different pre-processing techniques to the data to unlock the potential of CL (Zhang & Qi, 2005). Normalization and deseasonalization are the two main transformations applied to the data prior to importing them into the forecasting model itself. Additional pre-processing techniques for transforming (e.g., log transformation or Box–Cox transformation) and detrending (e.g., use of first differences or a linear trend) the series can also be considered, depending on the methods used (Ahmed et al., 2010; Makridakis, Spiliotis et al., 2018; Smyl, 2020). The underlying issue here is that, depending on the specific algorithms used for pre-processing, the results may vary. Therefore, selecting which method works best is entirely up to the practitioner. To further extend the previous point, CL methods require experienced forecasters to design and tune them appropriately. As opposed to their simpler counterparts, these methods have the capacity and are required to model significantly more information. As a result, they tend to be more complex, and designing them properly is more like an art, requiring knowledge and experience in fields not only directly relevant to forecasting, but also to programming and ML (Barker, 2020).

2.3. Remarks

Thus, traditional statistical methods are standard in forecasting, providing reliable predictions for almost any forecasting task. However, with the emergence of ML, there is value in exploring the concept of CL. From the ML standpoint, models trained in a series-by-series fashion are the most appropriate option when limited series are available for training. Provided that the series are adequately long and computational time is not a concern, ML methods can produce very accurate forecasts by capturing the specific patterns of a series. However, in many cases the series length will not be enough to properly train an ML model. Low-frequency data (e.g., yearly or monthly) and short-lived series dominate real-life forecasting applications. In addition, collecting more series is much easier than before, tipping the balance in favor of CL. Thus, even though it is more challenging to design an accurate CL forecasting method, unlocking the potential of ML methods and allowing them to acquire knowledge from large data sets is a promising way forward. Of course, apart from building accurate models trained across multiple series, looking into different ways to combine them with traditional ones may lead to better forecasting solutions (Smyl, 2020).

3. Experimental design

This section provides information about the data selected for performing the empirical evaluation of this study, the forecasting performance measures utilized, the benchmarks considered, and the forecasting framework used as a base for developing both traditional and CL forecasting methods.

3.1. Data

In order for the results of a study to be meaningful, they must be directly comparable to those of other relevant studies, as well as reproducible (Makridakis, Assimakopoulos et al., 2018). In this regard, we consider the data of the M4 competition, which is a well-known, established, and publicly available data set (Makridakis, Spiliotis et al., 2020).

For the sake of brevity, we consider only the monthly data of M4, which consist of 48,000 time series of six different domains and diverse characteristics (Spiliotis, Kouloumos et al., 2020). We retain the original design of the competition, thus keeping the last 18 observations of each series for testing (out of sample) and the remaining ones for training (in sample). As a result, in-sample data are used for hyper-parameter optimization and model training purposes, while out-of-sample data are used to measure the actual forecasting accuracy of the examined models.

3.2. Forecasting performance measures

In order to compare the accuracy of the forecasting methods examined in this study, we consider the official

accuracy measure of the M4 competition, i.e., the overall weighted average (OWA) of the symmetric mean absolute percentage error (sMAPE) (Makridakis, 1993) and the mean absolute scaled error (MASE) (Hyndman & Koehler, 2006). The OWA is defined as

$$OWA_{Model} = \frac{\frac{sMAPE_{Model}}{sMAPE_{Naive2}} + \frac{MASE_{Model}}{MASE_{Naive2}}}{2}, \tag{1}$$

$$sMAPE = \frac{2}{h}\sum_{t=n+1}^{n+h}\frac{|Y_t - F_t|}{|Y_t| + |F_t|} * 100\%, \tag{2}$$

$$MASE = \frac{\frac{1}{h}\sum_{t=n+1}^{n+h}|Y_t - F_t|}{\frac{1}{n-m}\sum_{t=m+1}^{n}|Y_t - Y_{t-m}|}, \tag{3}$$

where F_t is the forecast of the model at point t, Y_t the corresponding actual value of the series, h the forecasting horizon (i.e., 18), n the number of historical observations, and m the frequency of the series (i.e., 12 for monthly data). Note that, according to the M4 evaluation framework, the overall forecasting accuracy is computed by first estimating the MASE and sMAPE for each series individually and then averaging the derived values across all series, with the OWA computed on these global averages. The Naive 2 method was the primary benchmark of the M4 competition and is therefore used to scale the two measures. For more details about the OWA and the implementation of Naive 2, see Makridakis, Spiliotis et al. (2020).

However, despite the wide use of sMAPE and MASE in the forecasting literature, both measures, and especially sMAPE, have attracted criticism for their shortcomings (Goodwin & Lawton, 1999). Davydenko and Fildes (2016) discuss the limitations of these measures, arguing that sMAPE penalizes positive errors more than negative ones of the same absolute value, with MASE overestimating the performance of the Naive method used as a benchmark. The use of arithmetic means, instead of geometric ones, is also criticized by the authors, due to the vulnerability of arithmetic means to extreme values. As a result, the authors propose the use of the average relative mean absolute error (AvgRelMAE), a robust relative measure that suggests accuracy improvements over the benchmark used for scaling when its value is lower than one, and vice versa.

In order to make sure that the conclusions drawn in this study are representative, we use AvgRelMAE along with the OWA to evaluate the examined forecasting methods, using Naive 2 as a benchmark, as follows:

$$AvgRelMAE = exp(\frac{1}{N}\sum_{i=1}^{N} ln(\frac{MAE_{Model,i}}{MAE_{Naive2,i}})), \tag{4}$$

$$MAE = \frac{1}{h}\sum_{t=n+1}^{n+h}|Y_t - F_t|, \tag{5}$$

where N is the total number of series used in the evaluation.

Given that, apart from forecasting accuracy, method complexity (MC) and computational time (CT) are also important criteria for assessing the performance of different forecasting methods and evaluating the potential of new ones, we approximate these additional measures using some indicative variables as proxies: MC through the number of trainable parameters of the models used by the method, and CT through the time needed to sequentially train the models (in minutes). Although the CT can be improved by training the models in parallel, sequential model training on a single GPU was employed in an attempt to make the reported CT measures more interpretable. The CT was estimated using a system with the following characteristics: two Intel Xeon Gold 5118 CPUs @ 2.30 GHz, x64-based processor, 24 cores, 48 logical processors, and 64.00 GB of RAM, and two Quadro P5000 GPUs with 16 GB of RAM each.

3.3. Base forecasting framework

In order for the performance of the CL methods examined in this study to be directly comparable, we considered a base forecasting framework on which all the CL methods were based. Thus, any differences reported between the individual models are due to the different strategies used to extract information from the training sample, and not attributed to the special architectures or hyper-parameters of the underlying models. This framework includes two data pre-processing techniques and the main forecasting module.

3.3.1. Data pre-processing

In terms of pre-processing, two techniques are applied to the original series before using them as inputs to the forecasting module: seasonal adjustments and scaling.

The literature indicates that NNs typically struggle to capture the seasonal components of series (Barker, 2020; Nelson et al., 1999; Zhang & Qi, 2005). As a result, classical, multiplicative decomposition is used to seasonally adjust the series, as applied in the case of the Naive 2 benchmark (Makridakis, Spiliotis et al., 2020). According to this approach, only series that are characterized as seasonal by a 90% autocorrelation test are seasonally adjusted.

Note that we do not claim that NNs are unable to capture seasonal patterns, especially when large vectors are used as inputs and complex, deep NN architectures are considered (Oreshkin et al., 2019). However, given that the length of the M4 series is relatively short, and that their points are not chronologically aligned, identifying and matching seasonal patterns without using a seasonal adjustment technique is not trivial. Moreover, the aim of this study is to investigate the potential of different CL methods and not to explore complex NN architectures.

Data scaling is standard practice when it comes to developing NN forecasting models (Shanker et al., 1996) but becomes crucial for CL approaches where different series of different scales have to be imported into a single model. We consider the simple min–max transformation for scaling the series in a target range of [0, 1]. Scaling is applied to the seasonally adjusted series, provided that the original series is identified as seasonal. The inverse transformation can be applied to the derived forecasts, along with re-seasonalization if needed, to obtain the final forecasts.

3.3.2. Forecasting module

The forecasting module involves an ensemble of 50 NNs, given that the utilization of ensembles is considered beneficial for improving forecasting accuracy, mitigating uncertainties related with model initializations, and obtaining more robust and indicative results (Barrow et al., 2010; Kourentzes et al., 2014). The median operator is then used for aggregating the results of the NNs and obtaining the final forecasts.

The NNs used in the ensemble, to be called the base CL model, are simple feed-forward networks with fully connected layers, developed using Python 3.7.6, the Keras API v.2.3.1, and the Tensorflow v.2.1.0 as the back-end. Despite their relatively simple structure, tuning the architecture and defining the hyper-parameters of such NNs are complicated (Barker, 2020), and these choices have a great impact on the actual performance of the models (Bergstra et al., 2011).

In this respect, we define the optimal architecture and training hyper-parameters of the NNs by performing a hyper-parameter search. An ensemble of 50 NNs was built and trained for each combination of values and the forecasting accuracy of each ensemble was evaluated on an indicative validation set (the last 18 observations of the training set). For this task, the Tree-of-Parzen-Estimators (TPE) algorithm was used, implemented in the HyperOpt library for Python (Bergstra et al., 2015). The best set of values was then selected in order to build and train the models used to produce forecasts for the test set. The optimal hyper-parameter values determined by the search, along with the rest of the characteristics of the NNs, are presented in Table 1. It is important to point out that these optimal values are shared among all methods considered in this study. The architecture of the base CL forecasting model is also presented in Fig. 1.

Note that, in order to accelerate the training process, the most recent available values of each series are used as inputs to the NNs, both for training and extrapolation. Thus, instead of considering multiple windows for each series, we examine just the last window available per case. The length of the window is also determined by the hyper-parameter optimization process described above, to allow for possible long-term patterns and the remaining seasonal components to be learned.

3.4. Benchmarks

In addition to the Naive 2 method, used as a benchmark both for the OWA and the AvgRelMAE measures, we consider additional benchmarks so that the forecasting performance of the examined CL methods can be directly comparable to that of other CL methods examined in the literature, as well as of some traditional ones.

The first ML benchmark, to be named the MLP (multilayer perceptron), is an ensemble of 50 NNs trained in a series-by-series fashion using rolling windows of a single series as input (Makridakis, Spiliotis et al., 2018). Thus, 48,000 different MLP ensembles are trained to predict the series of the data set, each one specialized for a single series. In order to make the MLP benchmark more competitive, the architecture and training hyper-parameters of the individual models were optimized using the last in-sample window of the series and the TPE algorithm implemented in the HyperOpt library, as done in the case of the base CL forecasting model.

The second ML benchmark used in this study is the method originally submitted to the M4 competition by B. Trotta, ranked as the best "pure" ML entry of the competition (Makridakis, Spiliotis et al., 2020). According to this approach, the forecasts are produced by combining the outputs of several different NNs. First, two NN types are considered, the first trained on the monthly data and the second on the aggregated yearly series. The NN responsible for handling the monthly data consists of convolutional and fully connected layers, while the NN that handles the yearly series consists of only fully connected layers. Second, each model type has variations based on the number of inputs it receives. For example, 48, 120, and 240 data points are employed in the case of the monthly series. Third, different models are trained for each forecasting horizon. The submission of Trotta can be characterized as CL, since the models are trained across all the available series. For more information about this benchmark, see The M4 Team (2018).

Finally, additional established statistical methods that also served as benchmarks in the M4 competition were examined. The first benchmark is the simple exponential smoothing (SES) method (Hyndman et al., 2002), which is appropriate for predicting series without trends. The second benchmark is the simple, equally weighted combination (Comb) of SES, Holt, and Damped exponential smoothing. Comb was the main benchmark of the M4 competition and is appropriate for predicting series that display trends. The last benchmark considered is the Theta method, the top-ranked method in the M3 competition and the strongest official statistical benchmark in the M4 competition.

4. Feature-based methods

Exploiting time series features for improving forecasting accuracy is one of the most popular research topics in the forecasting community. Researchers and practitioners use such features in various ways, complementing or facilitating forecasting approaches that build on traditional forecasting models.

The most common applications that utilize time series features are those of model selection and combination. The main assumption made in these kinds of applications is that, depending on the particularities of the series, different methods are expected to perform well (Petropoulos et al., 2014; Spiliotis, Kouloumos et al., 2020). Armstrong and Collopy (2005) proposed a set of empirical rules for selecting the most appropriate forecasting method by observing the values of certain features. Nasiri Pour et al. (2008) introduced an FB NN model for demand forecasting. More recently, Talagala et al. (2018) introduced an automated FB method for predicting the series of the M4 competition where a random forest is trained over a large set of series, using time series features as input, to determine which method should be used for forecasting. In the area of model combination, Montero-Manso et al.

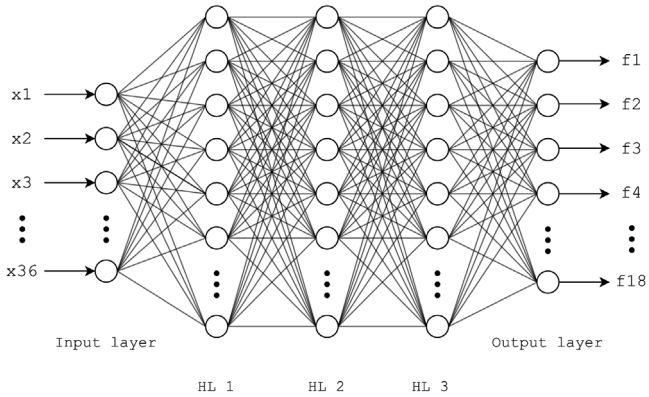

Fig. 1. The NN architecture of the base CL forecasting model.

Table 1
The architecture and the training hyper-parameters of the NNs used for the base CL forecasting model, along with the considered values for each hyper-parameter and the optimal values, as determined by the optimization performed over 100 iterations.

Hyper-parameter	Range of values considered	Optimal value
Input size (IS)	[12, 36, 48, 54]	36
Number of hidden layers	[2, 3]	3
Size of hidden layers	[0.5, 1.0, 1.5, 2.0] ×IS	1.5 × IS
Activation function	[Tanh, RelU, Leaky RelU]	RelU
Optimizer	[Adam]	Adam
Learning rate	[0.0001, 0.0005, 0.001, 0.005, 0.01]	0.001
Max training epochs	[500]	500
Early stopping patience	[2, 5, 10]	10
Batch size	[64, 128, 256, 512]	64
Loss function	[MAE, sMAPE]	MAE

(2020) utilized an FB algorithm to determine the optimal combination weights of several traditional forecasting models, reporting the second-best forecasting accuracy in M4.

Apart from complementing forecasting frameworks, time series characteristics are also valuable for analyzing the properties of large data sets. For instance, time series features are popular in applications related to clustering (Liao, 2005; Wang et al., 2006) and classifying series (Hatami et al., 2019; Nanopoulos et al., 2001) into different groups in an attempt to gather and process them in batches of similar ones, as well as for performing meaningful visualizations (Kang et al., 2017). Thus, time series features have become an important tool for reviewing different sets of series, assessing their diversity, and identifying patterns in an intuitive way.

The results of the M4 competition indicate that CL methods can be more competitive than their traditional counterparts, especially when it comes to ML ones (Makridakis, Spiliotis et al., 2020). However, after reviewing the various M4 submissions trained in a CL fashion, as well as the FB ones available in the literature, the question of whether a CL approach could benefit from the utilization of time series features along with the historical data becomes critical.

The answer to this question is rather straightforward if we consider how ML algorithms work. In order for an ML method to be effective, the training sample has to be dense (Barker, 2020). As noted previously, NNs do not assume an underlying DGP, meaning that a considerable amount of data of particular time series characteristics (e.g. trended and smooth series) must become available in order for the model to learn how to best capture and extrapolate series of similar patterns. Accordingly, if the characteristics of the series in the data set vary (e.g., stationary and noisy series exist along with trended and smooth series), more complex approaches may be required in the form of additional inputs, different training strategies, and different network architectures, in order for models to effectively handle the particularities of the series and produce accurate results. This is exactly the case for the M4 competition's data, which are characterized by high diversity and involve series of various patterns and, most importantly, of different densities per pattern (Spiliotis, Kouloumos et al., 2020). Training an NN across such a diverse, unbalanced data set could bias its forecast generation process, which, given the densities present, will work best for the most popular types of series in the training sample. In this regard, developing FB models capable of diversifying their forecasts based on the particularities of each series has great potential.

Apart from that, time series features are very effective in summarizing the basic, long-term characteristics of the series alone. Thus, providing such features as complements to NN models could further improve their performance by introducing information that otherwise would be extremely difficult to extract directly from the data points of the series. This is particularly relevant in the present study, where historical data are fed to the forecasting models in relatively short windows. On average,

the M4 monthly series consist of more than 200 data points, of which the 36 most recent are used to provide information only for the most recent series.

Before examining different alternatives for developing FB forecasting methods, another question has to be answered first: which features should be used in order to get the best out of such methods? There are literally dozens of potential features proposed in the literature that, depending on the situation, may either provide useful insights or mislead the models, letting them struggle to rationalize random patterns. Montero-Manso et al. (2020) consider 42 statistical time series features, while in different applications, features related to the date and time of the forecasts, as well as categorical features such as the series identifier, are also included in the forecasting framework (Salinas et al., 2020). For the sake of brevity, and since the series are seasonally adjusted beforehand, in this study we consider the following four statistical features, as proposed by Kang et al. (2017):

- The strength of the trend (measuring long-term changes at the mean level)
- Spectral entropy (measuring forecastability/randomness)
- First-order autocorrelation (measuring linearity)
- The optimal Box—Cox transformation parameter (measuring stability)

Using these four features as complements to the base forecasting module described in the previous section, we develop the following three FB methods:

- **CL-FB-D** *(features are used directly (D) as additional inputs to the base CL model)*: A straightforward way to exploit time series features is to use them as direct inputs to the base CL model along with the historical data, as presented in Fig. 2. The main hypothesis made here is that the last historical data points do not necessarily fully represent the long-term patterns of the series, which on the contrary are adequately summarized by the features. Thus, CL-FB-D, in contrast to the base CL model, will have the ability to extract more information from the data, become more specialized for different types of series, and therefore, derive more accurate forecasts.
- **CL-FB-C** *(features are used to cluster (C) the series, and a different base CL model is used for each cluster)*: A second approach to exploiting time series features is to use them to create clusters of series. Then, multiple NN models can be trained, each one specialized in forecasting the member series of a particular cluster. The intuition for this approach arises from the fact that modeling different types of series equally well requires a learning capacity that a relatively simple NN may not have. Thus, a reasonable way to address this issue is to build models that are specialized for predicting certain types of series. These models are still be trained across multiple series but are more focused in scope. In this regard, the four features are used as inputs to a k-means algorithm for the purpose of creating four clusters and assigning all the training series to them. The member series of each cluster are then used to train a cluster-specific model. When a series from the test set is to be forecasted, it is first assigned to a cluster and then extrapolated by the respective base CL model. By implementing this approach, we end up with four clusters of 20,619, 4,993, 14,642, and 7,746 series each.
- **CL-FB-E** *(features are used to populate the time series space equally (E), and the base CL model is used for forecasting)*: Another alternative is to balance the sample size of each type of series in the training sample. The intuition behind this approach arises from the lack of balance in the training set, which could lead the NN model to focus more on the most heavily populated types of series, thus deriving biased forecasts for the less frequently observed types of series. Thus, ideally, the NN should be trained using a diverse, yet balanced set of series. This strategy could be particularly useful in M4, as it consists of diverse and unequally populated types of series (Spiliotis, Kouloumos et al., 2020). As shown in Fig. 3, which provides a 2D visualization of the M4 monthly series in the feature space proposed by Kang et al. (2017), disproportionately more series can be found in certain areas of the feature space. In this regard, we split the feature space into four quadrants and then randomly exclude series from the most populated areas of the space in order for all the quadrants to contribute an equal number of series to the sample used for training. Once again, the base CL model is used for forecasting, trained with the balanced sample of the series created. By implementing this approach, we end up with 10,020 series for training the CL model.

5. Hybrid methods

Given that the accuracy of models trained in a CL fashion can be highly affected by the sample used for training, as well as the particular characteristics of the corresponding series, a reasonable way forward is to combine CL approaches with traditional models trained in a series-by-series fashion, so that both the information hidden in the complete data set and the particularities of the single series being extrapolated each time are taken into account.

The results of the M4 competition highlight the merits of such an approach. The top performing method, submitted by Smyl (2020), involved a truly hybrid method in the sense that series-specific methods (exponential smoothing) were combined concurrently with a CL dataset-specific component (a recurrent NN). The second-best performing method, submitted by Montero-Manso et al. (2020), involved multiple traditional models (both statistical and ML) utilized to generate the base forecasts, and an ML model (XGboost) trained across the entire data set to determine their combination weights.

Undoubtedly, there are numerous, vastly different alternatives for constructing hybrid forecasting methods, from simply averaging the forecasts produced by CL and traditional models to fully integrating them into a single

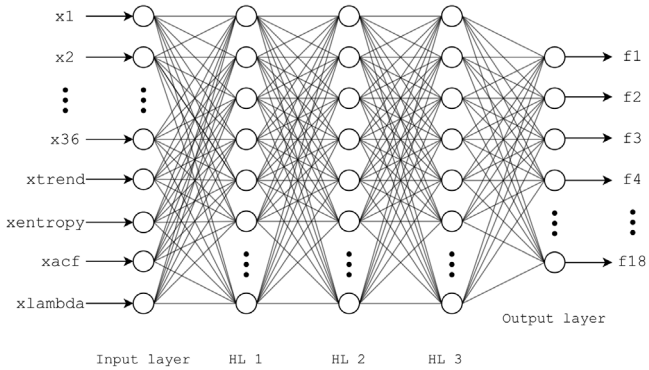

Fig. 2. The NN architecture of the CL-FB-D forecasting model.

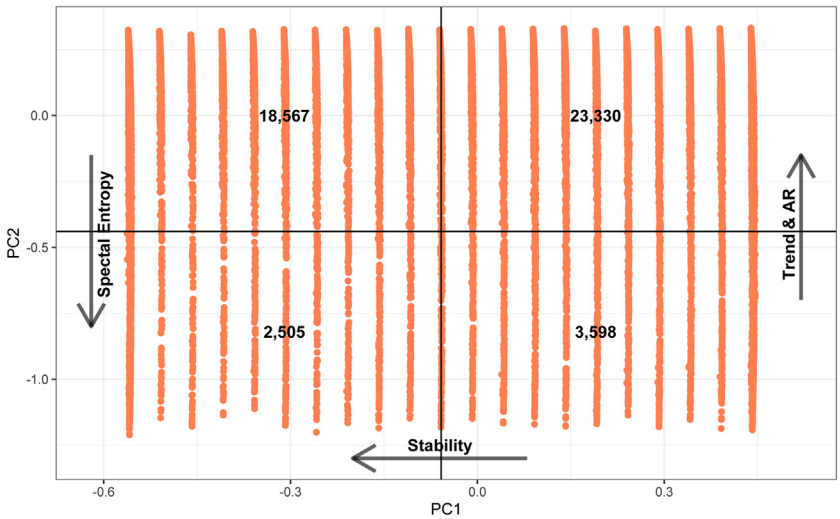

Fig. 3. The instance space of the M4 monthly series, generated using the methodological approach proposed by Kang et al. (2017) and the strength of the trend, spectral entropy, first-order autocorrelation, and optimal Box–Cox transformation parameter features as inputs. Following the forecasting approach of CL-FB-E, the space is divided into four quadrants, each one of a different sample of series.

method. For the sake of brevity, in this study we consider the following three relatively simple, yet indicative hybrid forecasting models, thus effectively investigating the potential of this kind of approach:

- **H-AVG** *(average of the base CL and Theta)*: This first approach averages the forecasts generated by the base CL model and the traditional Theta method used as a benchmark using equal weights, so that both the knowledge extracted from the entire data set and the specific characteristics of the series are taken into account.
- **H-LVL** *(the running level (LVL) of the series is provided as an additional input to the base CL model)*: In this approach, an alternative way of providing series-specific information to CL models is assessed. This kind of information is introduced by providing the base CL model with the running level of the series as input, along with the historical data of the series. Given that simple exponential smoothing (SES) (Gardner, 1985) is an excellent method for specifying the running level of a series, the last fitted value of SES is used for providing the base CL model with additional useful information, so that its forecasts are appropriately adjusted. The architecture of the H-LVL is presented in Fig. 4.
- **H-LVL-FB** *(the running level (LVL) of the series is provided as an additional input to CL-FB-D)*: This model is similar to H-LVL, but this time the running level of the series, as specified by SES, is provided as an additional input to the CL-FB-D model.

6. Results and discussion

The forecasting performance of the various CL approaches considered in this study, along with that of the established benchmarks, is summarized in Table 2. The second and third columns of the table present the forecasting accuracy of the examined models in terms of the OWA and AvgRelMAE, respectively. The last two columns show their MC and CT, respectively, to assess the trade-off between potential accuracy improvements and additional computational costs. It should be noted that

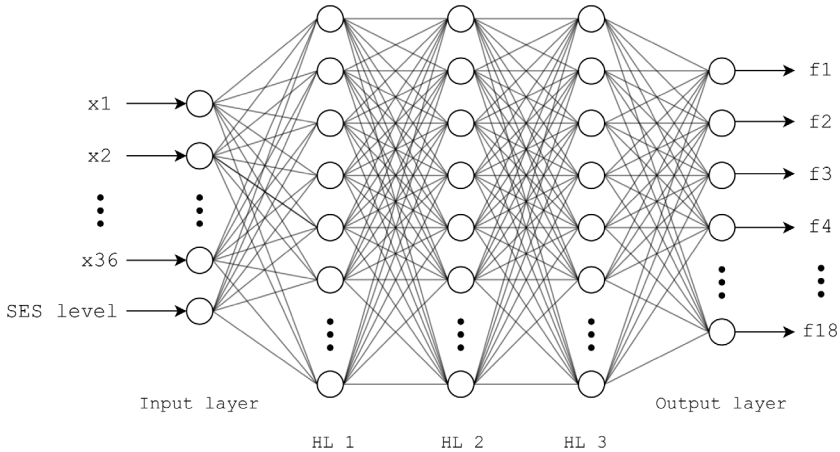

Fig. 4. The NN architecture of the H-LVL forecasting model.

Table 2
Forecasting performance across all 48,000 monthly series and 18 forecasting horizons of the various models considered in this study, along with that of the selected benchmarks. Model complexity (MC) refers to the number of trainable parameters, and computational time (CT) is measured in minutes.

Model	OWA	AvgRelMAE	MC	CT
Benchmarks				
SES	0.951	0.968	2	4
Comb	0.920	0.895	11	16
Theta	0.907	0.900	4	6
Best M4 ML submission	0.941	0.931	5,603	111
MLP	1.141	1.063	333	10,841
Base cross-learning model				
CL	0.899	0.873	8,928	76
Feature-based models				
CL-FB-D	0.887	0.861	10,878	75
CL-FB-C	0.900	0.874	8,928	103
CL-FB-E	0.908	0.887	8,928	18
Hybrid models				
H-AVG	**0.877**	**0.858**	8,932	82
H-LVL	0.897	0.873	9,258	66
H-LVL-FB	0.884	0.861	11,242	74

MC is a measure of how complex each method is, irrespective of the number of iterations needed for calculating forecasts for the entire data set. For example, SES, which has just two trainable parameters (viz., the initial value l_0 and the smoothing parameter a), is considerably simpler than base CL, despite the fact that these two parameters are calculated for all 48,000 series separately—resulting in 96,000 optimized parameters in total, as opposed to the 8,928 parameters for the base CL. The same principle holds true for all methods trained in a series-by-series fashion, as well as for the CL-FB-C, which trains models for each one of the four clusters.

The first major conclusion drawn by observing the results of Table 2 is that the utilization of CL leads to improvements in terms of forecasting accuracy. More specifically, the base CL model is found to be 22% more accurate than the MLP model, trained in a traditional series-by-series fashion, both according to the OWA and AvgRelMAE. Moreover, the improvements achieved by CL come with significantly less computational cost, due to the series-by-series training that traditional modeling requires. When compared against the more competitive statistical methods—viz., SES, Comb, and Theta—the base CL model is still more accurate, by 5%, 2%, and 1%, in terms of the OWA, and by 10%, 2%, and 3% in terms of the AvgRelMA, respectively. Yet, all three statistical benchmarks require significantly less time to complete the same forecasting task. Thus, it is confirmed that CL is a promising alternative to traditional time series forecasting methods, but at the cost of more computational time.

The second conclusion drawn is that the CL-FB-D model, which uses time series features as additional inputs to the base CL, is the most accurate FB approach out of the ones considered, and also the only FB model to outperform the base CL model, both in terms of the OWA and AvgRelMAE, by about 1.3%. The other FB models fail to outperform the base CL to a small degree, but are generally more accurate than the traditional benchmarks. Specifically, CL-FB-C and CL-FB-E provide more accurate forecasts, as indicated by both measures, with the exception of Theta, which is 0.13% more accurate than CL-FB-E in terms of the OWA. This could indicate that creating clusters of series and developing "specialists" for each of them is not the best way to apply CL. However, this could be attributed to the clustering method used by CL-FB-C, which is quite simplistic in nature, and to the significant reduction in sample size that the utilization of CL-FB-C and CL-FB-E models implies. For example, the four clusters created for training each specialist of CL-FB-C consist of 57% to 90% fewer series than the original data set used for training the base CL model. Similarly, the CL-FB-E model is trained using a sample of just 10,020 series, i.e., 79% fewer series than the base CL. In fact, given that CL-FB-E is slightly less accurate than the base CL, despite using significantly less data for training, it could be the case that focusing on the diversity of the data is equally as, or even more important than having a large number of series to work with.

In regards to the hybrid approaches considered, and before commenting on their performance, it is important

to note that in order for a combination to be meaningful, it must at least outperform both of its components. Our results indicate that H-AVG does provide more accurate forecasts than its individual components. It is 2.5% more accurate than CL and 3.3% more accurate than Theta in terms of OWA, while also being 1.7% more accurate than CL and 4.7% more accurate than Theta in terms of AvgRelMAE. Thus, H-AVG can be characterized as a meaningful and effective strategy for mixing traditional with CL forecasting methods. Moreover, H-LVL and H-LVL-FB slightly outperform the base CL and CL-FB-D, respectively, in terms of the OWA, while being equally as accurate in terms of the AvgRelMAE as their corresponding models that do not use the running level of the series. Although the accuracy improvements are marginal, our results suggest that there is some value in using the running level of the series as an additional input to a CL model.

Taken together, the results of Table 2 provide strong evidence that CL forecasting methods can be used effectively to accurately forecast numerous series in a timely manner. However, these findings pose a question about the size of the training set required for a CL model to achieve such an objective. In an attempt to answer this question, two additional experiments were conducted. First, an ensemble similar to the base CL model, to be named CL-0.25, was trained using 25% of all training samples, i.e. 12,000 series, which were randomly selected. Second, an additional ensemble, to be named CL-0.50, was trained using 50% of all training samples, i.e. 24,000 series, consisting of the same 12,000 series used by CL-0.25 and 12,000 more, also randomly selected. Both CL-0.50 and CL-0.25 were evaluated on their ability to forecast all 48,000 series. Table 3 summarizes the results for CL-0.25 and CL-0.50, along with those of the benchmarks used in the study. It is evident that the decrease of the training size leads to a decrease in the forecasting accuracy. CL-0.25 is 1.6% less accurate than the base CL, while CL-0.50 is about 0.5% less accurate than the base CL, according to the OWA and AvgRelMAE. However, when compared to the traditional methods, CL-0.25 and CL-0.50 are still competitive alternatives, with Theta being the only statistical method to provide slightly more accurate results than CL-0.25 in terms of the OWA. Given that CL-0.50 and CL-0.25 produced forecasts for the entire data set, including 24,000 and 36,000 series, respectively, that were not encountered during training, these results further encourage the use of CL.

There has been considerable discussion within the forecasting community about the merits of using ML methods for batch time series forecasting, especially for short and mostly unrelated series (Barker, 2020; Makridakis, Spiliotis et al., 2020). Some studies strongly support the utilization of such methods (Ahmed et al., 2010), while others question their advantages over standard traditional alternatives (Makridakis, Spiliotis et al., 2018). Our results indicate that ML methods do have strong potential over statistical ones, provided that CL is employed. More importantly, improvements in terms of forecasting accuracy can be achieved without the need for constructing complex CL models.

However, it is evident from the results of the M4 competition (Makridakis, Spiliotis, Assimakopoulos, 2018) that more sophisticated CL methods can further improve forecasting accuracy. The top performing method, submitted by Smyl (2020), employed a significantly more complex approach to combining series-specific components with components that hold information about common patterns from the whole data set. This submission is 4.7% more accurate than H-AVG, with an OWA score of 0.836 for the 48,000 monthly series. Similarly, the second-best method of M4, by Montero-Manso et al. (2020), considered significantly more time series features compared to this study, along with an advanced framework to exploit them for combining the forecasts of traditional methods. As a result, the submission of Montero-Manso et al. (2020) is 2.2% more accurate than H-AVG, with an OWA score of 0.858. Thus, the insights from the top-ranked submissions of M4, along with the findings of this study, should motivate systematic future research into developing more advanced CL models, as well as hybrids.

Table 3
Forecasting performance across all 48,000 monthly series and 18 forecasting horizons of the base CL architecture trained using subsets of the whole data set, along with that of the selected benchmarks. The computational time (CT), measured in minutes, is also reported.

Model	OWA	AvgRelMAE	CT
SES	0.951	0.968	4
Comb	0.920	0.895	16
Theta	0.907	0.900	6
Best M4 ML submission	0.941	0.931	111
MLP	1.141	1.063	10,841
Cross-learning models			
CL	0.899	0.873	76
CL-0.5	0.904	0.877	32
CL-0.25	0.913	0.887	18

7. Limitations & conclusions

This study investigated the potential of CL forecasting methods by comparing their accuracy to that of traditional models trained in a series-by-series fashion. Simple CL models, as well as FB and hybrid ones, were considered in an empirical evaluation to assess different ways of extracting information from large, diverse time series data sets and exploiting such information for forecasting.

Our results indicate that relatively simple CL methods can outperform traditional ones, such as SES, Comb, and Theta, as well as NNs trained in a series-by-series fashion. When time series features and other kinds of series-specific information are provided to the CL methods as input, their accuracy is improved further. In this regard, even simple combinations of CL methods and well-established statistical methods can be considered for improving forecasting performance. However, when compared to statistical methods, accuracy improvements come at much greater computational costs.

The main limitation of our study relates to its primary objective, which is to investigate the forecasting performance of CL methods: It is practically impossible for such an analysis to cover all possible CL methods and, therefore, although the conclusions drawn can be generalized, further research is needed regarding different CL

approaches. For example, the FB and hybrid approaches examined in our study, as well as the trained NNs, cannot be considered complex. As mentioned above, evidence from the top performing submissions (Montero-Manso et al., 2020; Smyl, 2020) in the M4 competition suggests that more sophisticated approaches result in more accurate forecasts. Thus, further advances towards more complex methods that share the same principles could potentially lead to even greater results, making the choice between traditional and CL methods more clear. A second limitation of our study relates to the data set used and whether it sufficiently represents series of different forecasting applications. Although the 48,000 monthly unrelated series of the M4 competition constitute a large and diverse data set with which to acquire representative results and draw trustworthy conclusions, further examination is required for different or more specialized areas of forecasting. For example, companies and organizations often deal with large sets of closely related and chronologically aligned series. Spiliotis, Makridakis et al. (2020) examined the performance of ML methods trained in both a CL and series-by-series fashion for the case of daily SKU demand forecasting. They concluded that although CL was beneficial in some cases, series-by-series training was more effective in others. Thus, further investigation is needed for such settings in order to determine the potential value of CL, allowing for shared trends, seasonal patterns, holidays, and special events to be captured effectively.

Despite the limitations discussed, the findings of our study can motivate future research in developing more advanced CL methods, as well as hybrids, as follows:

- Given that FB methods provided better results than simple CL ones, investigating the effect of adding more time series features as inputs to this kind of model is critical. More importantly, researchers should investigate the importance of such features and determine ways for selecting them in an optimal and automated way.
- Considering that optimally handling the particular characteristics of the series is probably more important than having large samples of series to train a model trained in a CL fashion, new and effective approaches should be developed to create more specialized CL forecasting methods. Clustering the series based on their particular characteristics and balancing the training sample are just some indicative ways of working in that direction.
- Since both traditional and CL approaches have their shortcomings, developing hybrid methods that combine their elements and exploit their benefits is crucial. Although challenging as a task, identifying best practices for integrated hybrid solutions is a promising way forward (Smyl, 2020).
- Given that this study considered relatively simple NNs for applying CL, more complex deep NNs should be constructed to take full advantage of the suggested approach. The work done in the field indicates such alternatives (Oreshkin et al., 2019).
- If NNs are capable of learning effectively from multiple series how to accurately predict individual ones in a CL fashion, similar approaches could be used for transferring this kind of knowledge to different data sets from the one used for training (Oreshkin et al., 2020). In this respect, generic, pre-trained models could become available for accurately predicting various types of series.
- In an era where COVID-19 has disrupted all aspects of everyday life, time series that capture human behavior and its impact are also affected. Thus, the task of forecasting and the subsequent decision-making process have become more complicated (Nikolopoulos et al., 2020). Although traditional forecasting frameworks can still provide useful forecasts in such settings (Petropoulos & Makridakis, 2020a), CL approaches may prove to be more appropriate by exploiting similar patterns observed across multiple countries and regions. Hence, further research on the use of CL methods in forecasting the evolution and the impact of COVID-19 could offer additional tools to practitioners and policymakers to deal with the current pandemic as well as future ones.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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